

5.33 Solution: (a) From Eq. (5.10),

$$\mathbf{F}_{12} = -\frac{\mu_0}{4\pi} I_1 I_2 \oint \oint \frac{(d\mathbf{l}_1 \cdot d\mathbf{l}_2) \mathbf{x}_{12}}{|\mathbf{x}_{12}|^3},$$

with $\mathbf{x}_{12} = \mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}$, and notice that

$$\nabla_R \left(\frac{1}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|} \right) = -\frac{\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|^3} = -\frac{\mathbf{x}_{12}}{|\mathbf{x}_{12}|^3},$$

the force can be expressed as

$$\mathbf{F}_{12} = I_1 I_2 \nabla_R \left(\frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|} \right) = I_1 I_2 \nabla_R M_{12}(\mathbf{R}),$$

with

$$M_{12}(\mathbf{R}) = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|}.$$

(b) Since

$$\nabla_R^2 \left(\frac{1}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|} \right) = -4\pi \delta(\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}),$$

and $R \neq |\mathbf{x}_2 - \mathbf{x}_1|$, unless the two loops touch each other, then

$$\nabla_R^2 \left(\frac{1}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|} \right) = 0,$$

and subsequently,

$$\nabla_R^2 M_{12}(\mathbf{R}) = 0.$$